

# Skills Summary

## Modelling Periodic Behaviour

Use this reference while completing your worksheet or when studying.

**The model:**  $y = A \sin(B(t - C)) + D$  (or cos).  $|A|$  amplitude,  $D$  midline,  $C$  horizontal shift, period  $= \frac{2\pi}{B}$ .

**Reading  $A$  and  $D$  off a graph** (max 20, min 6):  $A = \frac{20-6}{2} = 7$  and  $D = \frac{20+6}{2} = 13$ .

**Always radians** on this topic. A calculator in degree mode returns a plausible wrong number rather than an error, which is why it goes unnoticed.

### 1. Reading the period off a graph

A period is measured **peak to peak** (or trough to trough). A peak and the *next* trough are only **half** a period apart — doubling that gap is the step most often skipped. Once you have the period,  $B = \frac{2\pi}{\text{period}}$ .

**Example:** A curve has a maximum at  $x = 3$  and its next minimum at  $x = 9$ . Find the period.

**Step 1.** The two points given are a peak and the next trough, so the gap between them is half a period, not a whole one.

**Step 2.** The gap is  $9 - 3 = 6$ , so the period is  $2 \times 6$ .

$\Rightarrow$  **12**

**Try it:** A curve has a maximum at  $x = 2$  and its next minimum at  $x = 5$ . Find the period.

check: 9

### 2. Building a model from a situation

Turn the words into the four parameters, one at a time: the swing gives  $|A|$ , the centre gives  $D$ , “one cycle every  $P$  units” gives  $B = \frac{2\pi}{P}$ , and where the motion *starts* decides between sin, cos and a leading minus sign. Starting at the minimum is  $-\cos$ ; starting at the midline going up is  $+\sin$ .

**Example:** A wheel turns once every 10 minutes. What is  $B$  in its height model?

**Step 1.** Only the timing matters for  $B$ : one full turn is one period, so the period is the turn time and nothing else in the story is needed.

**Step 2.**  $B = \frac{2\pi}{10}$ , and the fraction reduces.

$\Rightarrow$   $\frac{\pi}{5}$

**Try it:** A wheel turns once every 8 minutes. What is  $B$ ?

check:  $\frac{\pi}{4}$

### 3. Evaluating a model

Substitute the time, simplify the angle to a familiar unit-circle value if you can, then finish the arithmetic. Reduce the angle *before* taking the sine:  $\frac{\pi(2)}{4}$  is  $\frac{\pi}{2}$ , and you know that one exactly.

**Example:**  $h(t) = 6 \sin\left(\frac{\pi t}{4}\right) + 10$ . Find  $h(2)$ .

**Step 1.** The input is given, so this is substitution — and the angle is in radians, so no degree conversion belongs anywhere.

**Step 2.**  $\frac{\pi(2)}{4} = \frac{\pi}{2}$  and  $\sin \frac{\pi}{2} = 1$ , so  $h(2) = 6(1) + 10$ .

$\Rightarrow$  **16** (the maximum: midline 10 plus amplitude 6)

**Try it:** Same  $h$ . Find  $h(6)$ .

check: 4

### 4. Solving a periodic equation on one cycle

Isolate the trig function first, then ask the unit circle for **every** angle in one revolution with that value — there are almost always two. Convert each angle back to a value of  $t$ , then keep the ones inside the given interval. One answer on a full cycle is a warning sign, not a result.

**Example:** Solve  $2 \sin\left(\frac{\pi t}{6}\right) = 1$  for  $0 \leq t < 12$ .

**Step 1.** Isolate the sine before touching the unit circle, or the reference angle comes out of the wrong equation.

**Step 2.**  $\sin\left(\frac{\pi t}{6}\right) = \frac{1}{2}$  at  $\frac{\pi t}{6} = \frac{\pi}{6}$  and at  $\frac{\pi t}{6} = \frac{5\pi}{6}$ , giving  $t = 1$  and  $t = 5$ .

$\Rightarrow$   **$t = 1$  and  $t = 5$**

**Try it:** Solve  $2 \cos\left(\frac{\pi t}{6}\right) = 1$  for  $0 \leq t < 12$ .

check: 10 and 8